Journal of Motor Behavior

ISSN: 0022-2895 (Print) 1940-1027 (Online) Journal homepage: www.tandfonline.com/journals/vjmb20

Extroversion-Related Differences in Speed of Premotor and Motor Processing as Revealed by Lateralized Readiness Potentials

Jutta Stahl & Thomas Rammsayer

To cite this article: Jutta Stahl & Thomas Rammsayer (2008) Extroversion-Related Differences in Speed of Premotor and Motor Processing as Revealed by Lateralized Readiness Potentials, Journal of Motor Behavior, 40:2, 143-154, DOI: 10.3200/JMBR.40.2.143-154

To link to this article: https://doi.org/10.3200/JMBR.40.2.143-154

Published online: 07 Aug 2010.

Submit your article to this journal

Article views: 234

View related articles

Citing articles: 2 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=vjmb20

Journal of Motor Behavior, 2008, Vol. 40, No. 2, 143–154
Copyright © 2008 Heldref Publications

Extroversion-Related Differences in Speed of Premotor and Motor Processing as Revealed by Lateralized Readiness Potentials

Jutta Stahl
Georg Elias Müller Institute for Psychology
University of Göttingen, Germany

Thomas Rammsayer
Institute for Psychology
University of Bern, Switzerland

ABSTRACT. To further elucidate extroversion-related differences in speed of sensorimotor processing, the authors obtained behavioral and psychophysiological measures as participants (16 introverts and 16 extroverts) performed a visual go/no-go task. Although no extroversion-related differences in reaction time emerged, introverts showed faster premotor processing but slower central and peripheral motor processing—as indicated by latencies of the lateralized readiness potential (LRP) and electromyographic (EMG) data, respectively—than extroverts did. Additional regression analyses revealed that stimulus-locked LRP latency, response-locked LRP latency, and N1 EMG amplitude accounted for 40% of overall variability in individual extroversion scores. On the basis of the present results, the authors introduce a compensation hypothesis that accounts for the common failure of researchers to demonstrate extroversion-related differences in reaction time. The present results challenge J. Brebner and C. Cooper's (1985) model of extroversion in which stimulus analysis is not slower in introverts than in extroverts. However, the present findings support the assumption of faster motor processing in extroverts.

Keywords: extroversion-related differences, lateralized readiness potential, psychophysical measures, sensorimotor processes

Although extroversion is a broad, multifaceted personality dimension (which includes traits such as sociability, talkativeness, and liveliness), experimental evidence of basic performance differences between introverts and extroverts in psychomotor tasks that require a simple motor response is also accumulating (Stelmack, 1985, 1997). The results of studies of extroversion-related individual differences in which investigators have used response time measures, such as a button press or release, present a rather inconsistent picture of positive, negative, or neutral results (for reviews, see Bullock & Gilliland, 1993; De Pascalis, 2004). A somewhat more consistent pattern of results is obtained when response latencies are decomposed into reaction time (RT) and movement time (MT). Researchers can accomplish the decomposition by using a response panel that comprises a home button and several target buttons (e.g., Jensen & Munro, 1979). RT is recorded as the time from stimulus onset to release of the home button, whereas MT is the time from release of the home button to the subsequent press of a target button. Although RT mainly reflects the latency of processes such as speed of stimulus recognition, stimulus evaluation, and response organization (Frewer & Hindmarch, 1988; Sanders, 1998; Smith, 1968), MT can be considered a valid indicator of speed of response execution (Frewer & Hindmarch; Jensen & Munro). In the majority of the few studies in which RT was measured independently of MT, researchers observed that MT, but not RT, is faster for extroverts than for introverts (Barratt, 1967; Doucet & Stelmack, 1997, 2000; Stelmack, Houlihan, & McGarry-Roberts, 1993; Wickett & Vernon, 2000). Those data suggest that response execution processes, as indexed by MT, are relevant to individual differences in extroversion.

In several electrophysiological studies, researchers have used event-related potentials (ERPs) to further elucidate differences in sensorimotor information processing between introverts and extroverts. In the vast majority of those studies, the researchers focused on N1 and P3 components of the ERP (for reviews, see De Pascalis, 2004; Stelmack, 1997; Stelmack & Houlihan, 1995). The N1 component is characterized by a negative deflection that peaks about 100 ms after stimulus onset. Investigators consider N1 to

Correspondence address: Jutta Stahl, Georg Elias Müller Institute for Psychology, University of Göttingen, Gosslerstrasse 14, D-37073 Göttingen, Germany. E-mail address: jstahl@uni-goettingen.de

be sensitive to stimulus characteristics such as intensity or modality and to variations of attention (Hillyard, Hink, Schwent, & Picton, 1973). The positive ERP deflection that peaks approximately 300 ms after stimulus onset, referred to as *P3*, is regarded as an indicator of processes associated with stimulus evaluation (e.g., Kutas, McCarthy, & Donchin, 1977; Pritchard, 1981) and response selection (e.g., Leuthold & Sommer, 1998; Verleger, 1997). Although in some studies, introverts' N1 amplitude was larger than that of extroverts (e.g., Doucet & Stelmack, 2000, Experiment 1; Rammsayer & Stahl, 2004), in others, no extroversion-related differences in N1 amplitude were found (e.g., Doucet & Stelmack, 2000, Experiment 2; Stahl & Rammsayer, 2004). A similar, inconsistent picture emerged for P3: Several investigators found larger P3 amplitudes for introverts than for extroverts (Cahill & Polich, 1992; Daruna, Karrer, & Rosen, 1985; Stenberg, 1994), but others obtained the opposite result (Ditraglia & Polich, 1991; Gurrera, O'Donnell, Nestor, Gainski, & McCarley, 2001) or failed to show any extroversion-related differences (Doucet & Stelmack, 2000; Stahl & Rammsayer).

Brebner and Cooper (1985) proposed an interesting model that has served as theoretical basis for investigating the dependence of information processing on extroversion. Brebner and Cooper differentiated between two major processing systems, *stimulus analysis* and *response organization*, each of which could adopt two types of states (excitation and inhibition). On the basis of their empirical findings, Brebner and Cooper (1974, 1978, 1985; Brebner & Flavel, 1978) hypothesized that introverts are characterized by a stronger excitation state as a result of stimulus analysis (S-excitation) than are extroverts. Those researchers postulated an opposite pattern for response organization, that is, that excitation resulting from response organization (R-excitation) is stronger in extroverts than in introverts. Brebner and Cooper (1985) concluded from their findings that extroverts prefer fast responding, whereas introverts are geared to introspection. Hence, those investigators postulated that introverts take more time for stimulus analysis to experience larger S-excitation, whereas extroverts are characterized by fast response organization to reach R-excitation. A more detailed review of the behavioral and electrophysiological studies on extroversion shows that the experimental tasks or settings varied widely across those studies. Different task requirements may differentially affect the two major systems, depending on extroversion- or introversion-related excitation states.

Because Brebner and Cooper's (1985) model raises several questions regarding the effects of extroversion on the temporal dynamics of processing, we used the lateralized readiness potential (LRP) for a further investigation in two previous studies (Rammsayer & Stahl, 2004; Stahl & Rammsayer, 2004). The LRP is derived from the readiness potential that appears several hundred milliseconds before voluntary hand movements and is larger contralateral to the hand to be moved. More specifically, in a choice-response task with the two hands, ERP amplitudes were found to be larger at scalp sites contralateral to the activated hand (Gratton, Coles, Sirevaag, Eriksen, & Donchin, 1988; Smid, Mulder, & Mulder, 1987). Thus, *LRP onset* indicates the beginning of hand-specific movement preparation and is defined as the point in time when the LRP waveform starts to significantly deviate from zero. The interval from stimulus onset to LRP onset is referred to as *stimulus-locked LRP* (*S-LRP*) *latency*, whereas the interval between LRP onset and completion of the overt motor response is defined as *response-locked LRP* (*LRP-R*) *latency*. S-LRP latency reflects the duration of perceptual and cognitive processes that occur before central response activation. On the other hand, LRP-R latency indicates the time required for central response organization and execution of the overt motor response (Osman & Moore, 1993). On the basis of those considerations, researchers believe that LRP onset separates premotor from motor processes and that S-LRP latency represents an index of speed of premotor processing and LRP-R is a measure of motor-processing time.

Rammsayer and Stahl (2004) obtained LRPs in an auditory two-choice go/no-go task to examine speed of premotor (stimulus analysis) and motor (response organization) processing in introverts and extroverts. With that task, they found longer LRP-R latencies for introverts than for extroverts, indicating faster speed of motor processing in extroverts than in introverts. However, no extroversion-related differences could be shown for S-LRP latencies. Investigators have attributed the failure to demonstrate a difference in S-LRP latencies to the low task demands induced by the auditory pitch-discrimination task (go signals = 800 and 1200 Hz; no-go signal = white noise). Therefore, in a second study, Stahl and Rammsayer (2004) applied a two-choice letter-discrimination task to increase premotor (i.e., cognitive) task demands and to change the stimulus modality. In that study, they found that S-LRP latencies were shorter for introverts than for extroverts, indicating faster premotor information processing in introverts than in extroverts. That finding challenged Brebner and Cooper (1985), whose model suggests that there is prolonged stimulus analysis in introverts (for a detailed consideration of that issue, see the present Discussion section). Unfortunately, the two groups no longer differed in LRP-R latencies. Stahl and Rammsayer suggested that their failure to demonstrate an extroversion-related difference in speed of motor processing resulted from the fact that their letter-discrimination task did not include a no-go condition. We therefore designed the present study to obtain converging evidence that introverts and extroverts are differentially susceptible to specific task demands associated with premotor and motor aspects of sensorimotor information processing. For that purpose, we used a visual two-choice go/no-go reaction task combining the task demands predicted to be crucial for identifying extroversion-related differences in speed of premotor and motor processing, respectively.

Because of Stahl and Rammsayer's (2004) findings in a similar visual letter-discrimination task, in the present study we expected to find shorter S-LRP latencies for introverts than for extroverts, whereas we predicted shorter LRP-R latencies for extroverts on the basis of Rammsayer and Stahl's (2004) findings in a go/no-go task. Furthermore, to obtain separate measures for speed of central and peripheral motor processing, we recorded electromyographic (EMG) activity in addition to ERP waveforms (N1 and P3 components) and behavioral measures (RT and response force [RF]). Because of the inconsistent empirical findings with respect to effects of extroversion on those measures, we made no explicit predictions.

Method

Participants

We screened the personalities of 56 first-year female undergraduate students by having them fill in Ruch's (1999) German adaptation of the extroversion and neuroticism scales of the Eysenck Personality Questionnaire-Revised (EPQ-R; Eysenck & Eysenck, 1991). To control for well-known gender differences in evoked potentials (e.g., Cahill & Polich, 1992) and extroversion (e.g., Lynn & Martin, 1997), we used only women. The extroversion and neuroticism scales consist of 23 and 25 items, respectively. In a first step, we divided participants into two groups on the basis of below- and above-median scores on extroversion. We considered participants with extroversion scores less than 16 to be *introverts* and participants with extroversion scores greater than 16 to be *extroverts*. We excluded participants who scored 16. Then, because low scores on extroversion tend to be positively associated with neuroticism (Eysenck, 1967), we matched introverted and extroverted participants. This procedure resulted in 16 female undergraduate introverts and 16 female undergraduate extroverts. Respective mean extroversion and neuroticism scores were 9.7 and 13.4 for the introvert group and 17.9 and 12.9 for the extrovert group. Participants ranged in age from 19 to 41 years ($M \pm SEM = 22.2 \pm 0.9$ years), were right-handed, nonsmokers, and had normal hearing and normal or corrected-to-normal vision. They received course credit for their participation in the study and were naive about our experimental hypotheses.

Behavioral Measures

Apparatus and stimuli. We assessed RF and RT by using force-sensitive keys (cf. Giray & Ulrich, 1993). Each force key was composed of a leaf spring (110 mm × 19 mm × 2 mm) held by an adjustable clamp at one end while the participant pressed the other, free end with her index finger. A force of 1000 cN bent the free end of the leaf spring by approximately 2 mm. Each response key was mounted on a board, providing full forearm support. Furthermore, a chin rest maintained a constant posture and 65-cm distance to the computer screen.

We attached strain gauges near the leaf spring's fixed end. Thus, when the participant pressed the key, the strain gauges produced an analog electrical signal corresponding to the force applied to the leaf spring. We digitized and recorded the signal for 2,000 ms with a sampling rate of 500 Hz, starting at the onset of the visual response signal. For visual response signals, we presented the letters *u*, *v*, and *w* for 1,000 ms in the center of the computer screen; from the eye, the letters on the screen subtended a visual angle of about 0.60°.

Procedure. The letters *u* and *v* served as go signals. We assigned each letter to a right-hand or a left-hand response. For each participant, we held the letter assigned to the hand constant, but we balanced the assigned letters across participants. Participants used one response key for each hand. They responded with a brief flexion of either the right or the left index finger. We instructed participants to respond as fast as possible to the go signals but to withhold a response in case the no-go signal *w* occurred. To account for the speed–accuracy tradeoff, we asked them to respond fast, but without making too many errors. An experimental session consisted of six blocks. The first block served as a practice block and ensured that the participants were familiar with the task and the assignment. Five experimental blocks followed the practice block. Each block consisted of 12 left-hand, 12 right-hand, and 12 no-go trials. The participant initiated the next block when she felt ready to continue the experiment. Time between blocks averaged about 1 min across all participants. We randomized the order of trials within each block. The session lasted approximately 20 min.

During the practice block, we provided visual feedback after each trial. However, we presented no such feedback for the five experimental blocks. At the beginning of each trial, we presented a white fixation cross for 500 ms as a ready signal. The duration between the ready signal and the onset of the visual response signal corresponded to the sum of a constant period of 500 ms and an exponentially distributed random variable with a mean of 2,000 ms. That duration ensured that anticipatory responses were minimal.

We determined mean RT and mean peak RF for each type of trial across the five experimental blocks. Although we defined *RT* as the first moment at which RF exceeded a criterion of 50 cN after onset of the visual response signal, peak RF represented the maximum force value recorded during a given trial. As an additional indicator of response dynamics, we determined time to peak RF (TTP). *TTP* refers to the temporal interval from onset of the overt motor response to the point in time when participants reached peak RF. We based all behavioral data on correct responses. We considered wrong-hand responses, misses, RTs greater than 1,000 ms, and no-go errors as response errors, and we therefore excluded them from them data analysis.

Electrophysiological Recordings

We recorded the electroencephalogram (EEG) with silver chloride electrodes at five scalp loci referenced to the right mastoid. We selected scalp sites F*z*, C*z*, and P*z* according to the standard international 10–20 system (Jasper, 1958).

We positioned sites C3′ and C4′ 1 cm anterior to sites C3 and C4, respectively. We chose those positions because of their correspondence to the hand areas of the precentral motor cortex (cf. Coles, 1989). We recorded horizontal and vertical movements in the electrooculogram (EOG) from electrode positions supra- and infraorbital to the right eye and 2 cm external to the outer canthus of each eye. We fixed the active reference electrode on the left mastoid and the passive electrode on the right mastoid. In addition, we obtained EMG activity by using bipolar recordings from the ventral forearms. To determine the electrode positions, we trisected the wrist–elbow distance. We located the first electrode position at the first third (taken from elbow), and we positioned the second electrode at 5 cm ventral to the first position (Fridlund & Cacioppo, 1986).

Using a Neuroscan (Charlotte, NC) data-acquisition unit, we recorded the EEG continuously with a sampling rate of 1000 Hz. The time constant was infinite, and we set a low-pass filter to 70 Hz for all channels. We epoched the EEG offline into epochs that lasted from 100 ms before onset of the visual response signal until 900 ms after signal offset for the stimulus-locked EEG analyses. The 100-ms interval preceding the onset of the response signal served as a reference for baseline correction. We rectified the EMG offline. For response-locked averaging of EEG and EMG, we used an epoch ranging from 250 ms before participants reached the response criterion of 50 cN until 300 ms after then. We used a 100-ms interval starting 700 ms before participants reached the response criterion as the baseline reference for the response-locked data analyses. We screened all data for EOG artifacts, and we rejected contaminated trials exceeding a maximum-to-minimum amplitude of ±70 μV in any of the two EOG channels. We digitally rereferenced data to linked mastoids.

We averaged artifact-free single-trial EEG waveforms separately for each participant, each experimental condition, and each electrode position to obtain individual ERP waveforms across experimental blocks. The mean number of trials included in individual average waveforms was 98 (range = 91–119 trials). Separately for electrode sites F*z*, C*z*, and P*z*, we obtained individual N1 amplitudes from individual averaged waveforms as the maximum negative peak within an (80–180-ms) interval following onset of the response signal. We determined P3 amplitude as the maximum positive peak relative to baseline within a time window ranging from 300 ms to 600 ms after stimulus onset. We scored the N1 and P3 latencies at the mean maximum amplitude of the N1 and P3 components, respectively.

We recorded LRP waveforms from electrodes located above the left and right motor cortices (C4′ and C3′, respectively), and we calculated the LRP as suggested by Coles (1989). We computed difference waveforms between the two electrode sites as a function of the responding hand in a given trial. On each trial, we subtracted the recording from the ipsilateral hemisphere from contralateral recordings. Thus, on a trial requiring a left-hand response, we computed the difference wave C4′ minus C3′, whereas we computed C3′ minus C4′ for right-hand trials. In that procedure, all electrical brain activity distributed symmetrically across the scalp is eliminated. We then averaged the single-trial difference waveforms separately for left-hand and right-hand trials. In a last step, we computed the LRP as the mean of the average difference waves. Deviations of the resulting LRP from zero toward increased negativity indicated activation of the correct response hand at the level of the motor cortex.

To increase the signal-to-noise ratio before onset detection, we determined S-LRP and LRP-R onset latencies by using a jackknife-based scoring method with a relative criterion of 50% of LRP peak amplitude (cf. Miller, Patterson, & Ulrich, 1998; Stahl & Gibbons, 2004). Using jackknifing, one can determine a participant's LRP onset score from the averaged LRP waveform by including the LRP waveforms of all participants except that participant's. We performed the procedure separately for each participant.

We defined *EMG onset* as the point in time when EMG activity deviated by 10% of maximum EMG amplitude from baseline (Osman, Moore, & Ulrich, 1995). We defined *stimulus-locked EMG* (S-EMG) *latency* as the temporal interval from stimulus onset to S-EMG onset. We defined *response-locked EMG* (EMG-R) *latency* as the temporal interval from EMG-R onset to the key press. We determined S-EMG amplitude as the maximum positive peak within an interval from 100 ms to 800 ms after stimulus onset. We measured EMG-R amplitude as the maximum positive peak within an interval starting 100 ms before EMG-R reached the 50-cN response criterion and ending 100 ms after then (see the foregoing comments).

To test the predictions made for S-LRP latency (introverts' would be shorter than extroverts') and LRP-R latency (introverts' would be longer than extroverts'), we applied one-tailed *t* tests. We investigated differences between introverts and extroverts in the remaining dependent variables by using two-tailed *t* tests.

All reported correlation coefficients are Pearson correlations. Furthermore, we performed a stepwise multiple regression analysis to identify powerful dependent variables that predicted individual extroversion scores.

Results

Behavioral Data

For all go trials, about 0.1% were wrong-hand responses, less than 0.1% were misses, and 0.1% of responses were slower than 1,000 ms. Participants made about 1.5% no-go errors. A significant difference between introverts and extroverts could be established only for no-go errors, $t(30) = 2.35$, $p < .05$; for all other comparisons, $t(30) < 1$. Extroverts (1.98%) responded more frequently on no-go trials than did introverts (0.97%).

Extroversion did not reliably affect response speed, $t(30) = .42$, $p > .1$; mean RT of introverts was 521 ms, and mean RT of extroverts was 527 ms. Mean difference in peak RF between

the two groups was significant, $t(30) = 1.89$, $p < .05$; responses of extroverts (506 cN) were stronger than were responses of introverts (320 cN). No significant TTP difference was obtained, $t(30) = 1.12$, $p > .1$; mean TTPs were 104 ms and 119 ms for introverts and extroverts, respectively.

Electrophysiological Data

Figure 1 depicts the stimulus-locked grand-average ERP waveforms as a function of extroversion. Visual inspection of the grand average shows that the N1 component was most prominent at the F*z* electrode site, whereas the P3 component was more pronounced at P*z*, as suggested by previous findings (e.g., Doucet & Stelmack, 2000). Therefore, we used F*z* and P*z* electrode sites for analyzing extroversion-related differences in N1 and P3 components, respectively. We were unable to confirm a significant effect of extroversion on either N1 latency, $t(30) < 1$, or N1 amplitude, $t(30) < 1$. For P3 amplitude, there was also no indication of a significant effect of extroversion, $t(30) < 1$. Furthermore, we found no significant difference in P3 latency between introverts and extroverts, $t(30) < 1$. See Table 1 for all comparisons.

Figure 2 shows the grand-average LRP waveforms for introverts and extroverts. Because the jackknife procedure strongly reduces the standard errors of means, we corrected the obtained *t* values according to the procedure described

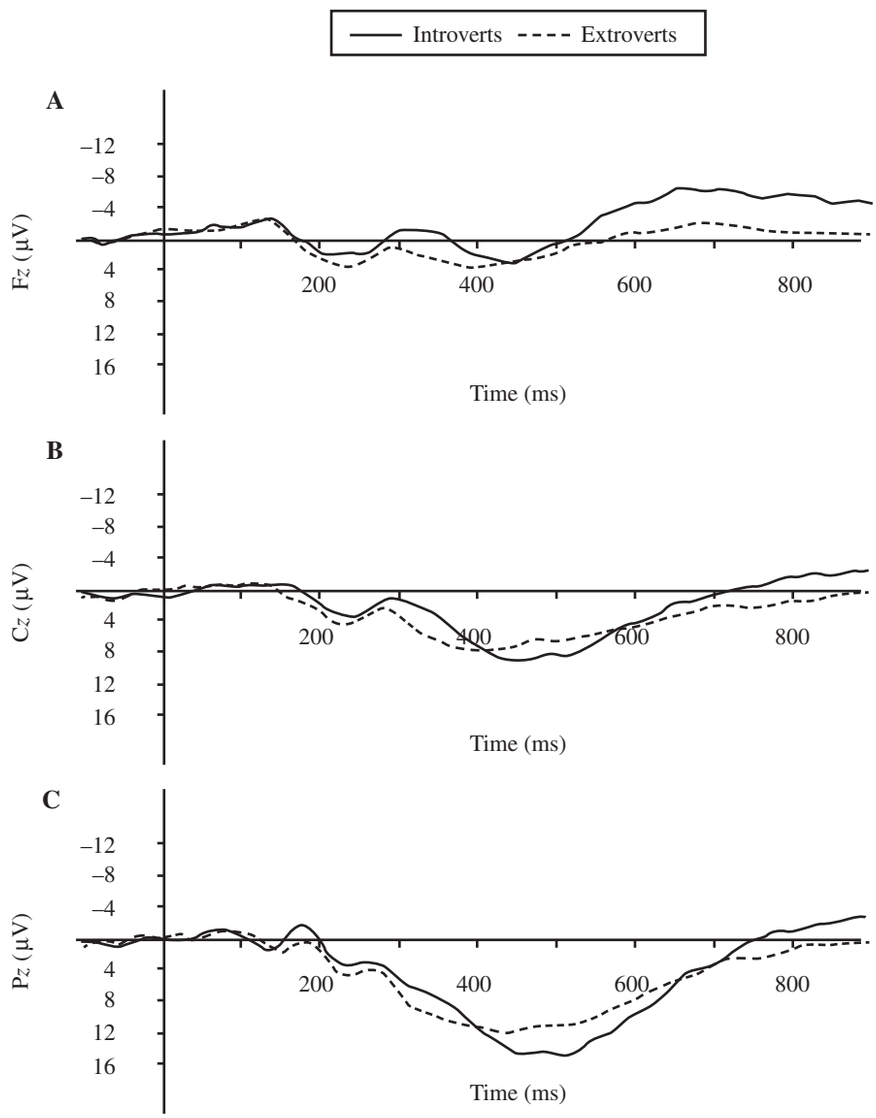

FIGURE 1. Grand-average waveforms of introverts and extroverts for (**A**) F*z* (midline frontal electrode site), (**B**) C*z* (midline central electrode site), and (**C**) P*z* (midline parietal electrode site).

TABLE 1. Means and Standard Errors of the Mean (*SEM*) of All Behavioral Data Measures in Introverts and Extroverts

	Introverts		Extraverts			
Measure	*M*	*SEM*	*M*	*SEM*	*t*	*d*
No-go error (%)	0.97	0.23	1.98	0.36	2.35**	0.84
Reaction time (ms)	521	7.6	527	14.2	0.42	0.15
Response force (cN)	320	39.7	506	90.3	1.89	0.69
Time-to-peak (ms)	104	8.1	119	11.0	1.12	0.40
N1 amplitude (μV)	–3.3	0.53	–2.9	0.31	0.61	0.22
N1 latency (ms)	126	5.3	130	3.3	0.63	0.23
P3 amplitude (μV)	14.7	1.5	12.9	1.6	0.80	0.29
P3 latency (ms)	513	12.4	506	9.8	0.49	0.18
S-LRP latency (ms)	365	6.6	389	10.9	2.15*	0.77
LRP-R latency (ms)	106	9.0	87	9.3	1.92*	0.69
S-EMG latency (ms)	452	8.3	454	7.4	0.67	0.24
S-EMG amplitude (μV)	9.6	5.7	9.0	4.3	0.33	0.11
EMG latency (ms)	48	9.1	30	3.6	1.93	0.69
EMG amplitude (μV)	22.0	5.9	20.0	3.8	0.33	0.11

Note. *d* = effect size (Cohen, 1988). N1 and P3 are measures of event-related potential (ERP). S-LRP and LRP-R = stimulus-locked and response-locked lateralized readiness potential, respectively. S-EMG and R-EMG = stimulus-locked and response-locked electromyogram, respectively.
$^{*}p < .05$ (one-tailed). $^{**}p < .05$ (two-tailed).

by Miller et al. (1998). A significant mean difference in S-LRP latency between extroverts and introverts was revealed, $t(30) = 2.15$, $p < .05$, one-tailed. Mean S-LRP latencies were 365 ms and 389 ms for introverts and extroverts, respectively. That finding indicates faster stimulus processing in introverts than in extroverts. Mean difference in LRP-R latency between introverts and extroverts also reached statistical significance, $t(30) = 1.92$, $p < .05$, one-tailed. Mean LRP-R latencies were 106 ms and 87 ms for introverts and extroverts, respectively, suggesting faster central and peripheral motor processing in extroverts than in introverts.

The grand-average stimulus-locked and response-locked EMG waveforms of introverts and extroverts are presented in Figure 3. We were unable to confirm a reliable difference for either S-EMG latency, $t(30) = 0.24$, $p > .1$, or S-EMG amplitude, $t(30) = 0.14$, $p > .1$. However, mean R-EMG latency tended to be longer in introverts (48 ms) than in extroverts (30 ms), $t(30) = 1.91$, $p < .07$. No reliable mean difference between the two groups was obtained for EMG amplitude, $t(30) < 1$ (see Table 1).

Correlational Analyses

Researchers have often criticized results based on median split of scale scores on the ground that the respective group means are somewhat arbitrary. Therefore, we also analyzed the present data by using a correlational approach, treating extroversion as a continuous variable. Researchers can obtain the correlation between two (nonjackknifed) variables from the correlation between jackknifed variables (e.g., LRP and EMG latencies) and nonjackknifed variables (e.g., extroversion scores, RTs, or ERP measures) by simply inverting the signs (cf. Stahl & Gibbons, 2004). All reported correlation coefficients are appropriately inverted. It should also be noted that the jackknifing procedure does not affect the significance level of the correlation coefficients. The interrelations between individual extroversion scores, behavioral performance, and electrophysiological measures are presented in Table 2. Based on a median split, a significant positive correlation between extroversion score and S-LRP latency, $r = .36$, $p < .05$, and a reliable negative correlation between extroversion score and LRP-R latency, $r = -.38$, $p < .05$, corroborated the aforementioned results. Higher levels of extroversion were associated with longer S-LRP latencies but shorter LRP-R latencies. The correlation between extroversion and N1 amplitude just failed to reach the 5% level of statistical significance, $r = .34$, $p < .07$.

To further elucidate the relationship between extroversion and the obtained dependent variables, we performed stepwise multiple regression analyses (see Table 3). We identified the most powerful predictor of the psychometrically assessed level of extroversion, LRP-R latency, which accounted for 14.5% of the total variance of overall variability in extroversion. When we combined LRP-R latency and S-LRP latency as two predictors of extroversion, we were able to explain 25.4% of the total variance. The combined effect of LRP-R and S-LRP latencies yielded a statistically significant increase in explained variance of $R^2 = 10.9\%$, whereas LRP-R alone accounted for 14.5% of the explained variance, $F(1, 29) = 4.2$, $p < .05$. Eventually, after we included N1 amplitude as a third predictor variable in addition to LRP-R and S-LRP latencies, the portion of explained variance of variability in

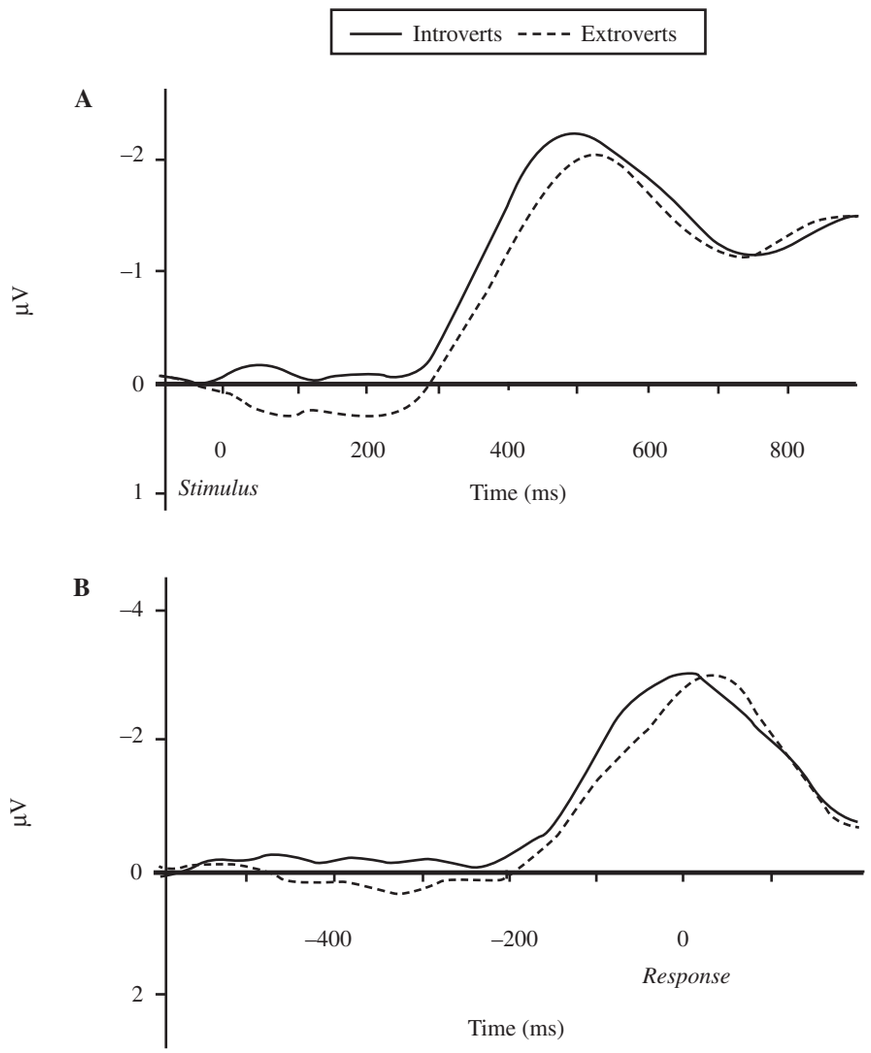

FIGURE 2. (**A**) Stimulus-locked and (**B**) response-locked lateralized readiness potentials of introverts and extroverts.

extroversion increased to 40.3%. The combination of those three predictor variables resulted in an additional significant increment in R^2 of 15.0% compared with the R^2s of LRP-R and S-LRP latencies as predictor variables, $F(1, 28) = 7.2$, $p < .05$. Inclusion of any other dependent variable in the regression model did not provide a further reliable increase in explained variance.

Discussion

Our aim in the present study was to investigate extroversion-related differences in speed of premotor and motor processing by means of a visual, two-choice go/no-go response task. We recorded RT, peak RF, TTP, and response errors as behavioral measures. In addition, we obtained electrophysiological measures N1 amplitude, N1 latency, P3 amplitude, P3 latency, S-LRP latency, LRP-R latency, EMG latency, and EMG amplitude. With regard to behavioral measures, we observed no extroversion-related difference in RT. That finding is in line with the results reported in numerous previous studies (e.g., Brocke & Liepmann, 1985; Keuss & Orlebeke, 1977; Rammsayer & Stahl, 2004; Stahl & Rammsayer, 2004; Wickett & Vernon, 2000). However, extroverts made reliably more no-go errors than did introverts. That finding is consistent with previous studies (Brebner & Flavel, 1978; Rammsayer & Stahl) that also yielded a significantly higher rate of errors of commission and no-go errors for extroverts than for introverts. Extroverts' higher rate of no-go errors provides converging evidence for Brebner and Cooper's (1985) assumption that extroverts should have more difficulty withholding a motor response than should introverts (Brebner & Flavel; Helmers, Young, & Pihl, 1997).

The electrophysiological analyses of both amplitude and latency of traditional ERP components (N1 and P3) could not demonstrate reliable extroversion-related differences.

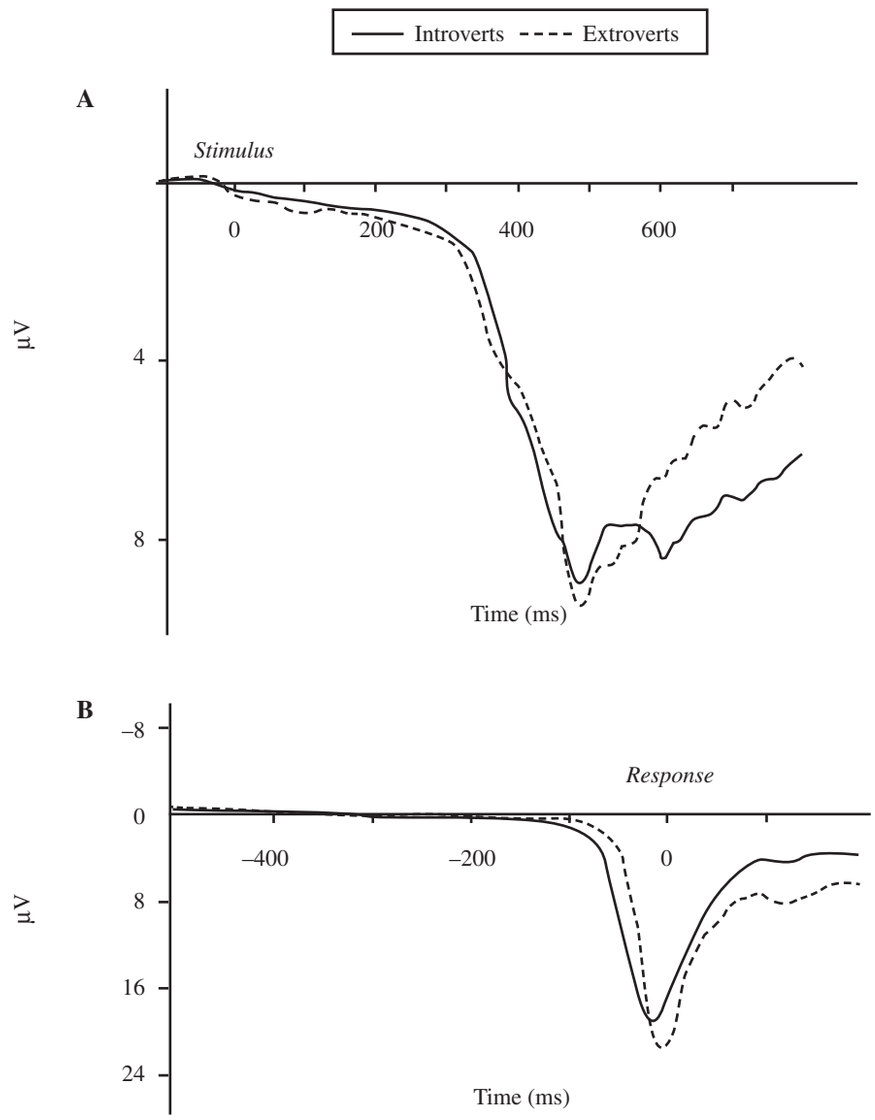

FIGURE 3. (**A**) Stimulus-locked and (**B**) response-locked electromyograms of introverts and extroverts.

That finding accords with several previous ERP studies in which researchers used visual choice RT tasks (e.g., Doucet & Stelmack, 2000; Stahl & Rammsayer, 2004). The psychophysiological LRP analyses that we applied to experimentally dissociate speed of premotor and motor processing yielded the most promising finding. Introverts showed reliably shorter S-LRP latencies compared with those of extroverts. Furthermore, correlational analyses yielded a positive relationship between individual extroversion scores and S-LRP latencies. Those findings were indicative of faster speed of premotor information processing in introverts than in extroverts. Such a conclusion is consistent with the outcome of a preceding LRP study in which Stahl and Rammsayer used a similar letter-discrimination task. At first glance, however, both findings challenge Brebner and Cooper's (1985) assumption that stimulus analysis is longer in in introverts than in extroverts, which they based on their 1978 findings. Nevertheless, task demands of the latter experiment were rather different from those of the present one. For example, Brebner and Cooper (1978) did not instruct their participants to respond as fast as possible; instead they invited participants to inspect the stimulus material (e.g., dot patterns, slides of streetscapes or architecture) without any time pressure and to press a button to start the next stimulus presentation. Introverts inspected fewer slides than did extroverts in the same amount of time. The present results and Stahl and Rammsayer's findings demonstrate that, if required, introverts are able to

TABLE 2. Correlations Between Extroversion, Behavioral Measures, and Electrophysiological Measures

Measure	Extraversion	E%	RT	RF	TTP	S-LRP	LRP-R	SE L	SE A	RE L	RE A	N1 L	N1 A	P3 L
Percentage of no-go errors (E%)	.34													
Response time (RT)	.01	.14												
Response force (RF)	.30	.34	.19											
Time-to-peak (TTP)	.24	.40*	.24	.86***										
S-LRP latency (S-LRP)[a]	**.36***	.35*	.31	.22	.24									
LRP-R latency (LRP-R)[a]	**–.38***	–.04	–.24	–.12	–.09	**–.07**								
S-EMG latency (SE L)[a]	.10	.14	.21	.19	.07	.32	.17							
S-EMG amplitude (SE A)[a]	.04	.11	.19	.21	.31	.21	.13	.50**						
EMG-R latency (RE L)[a]	–.33	.06	–.17	–.21	–.06	.08	–.11	.13	–.05					
EMG-R amplitude (RE A)[a]	–.02	–.24	–.16	.25	.32	.40*	.23	.11	.39*	.59**				
N1 latency (N1 L)	.14	–.16	–.04	–.26	–.24	–.06	–.40*	–.06	–.03	.01	–.35			
N1 amplitude (N1 A)	**.34**	.05	.01	.31	.32	**–.17**	**–.03**	–.17	–.21	–.22	.08	–.22		
P3 latency (P3 L)	–.01	–.18	.39*	.32	.26	.07	.14	.07	.09	.05	.09	–.28	.16	
P3 amplitude (P3 A)	–.11	.04	–.36	–.18	–.04	–.19	.08	–.19	.09	–.09	.13	–.01	.18	–.27

[a]Sign of jackknife-based coefficients was changed following the recommendation of Stahl and Gibbons (2004). Coefficients in bold type are correlations between the variables that were included in the regression analysis (see Table 3).
*$p < .05$ (two-tailed). **$p < .01$ (two-tailed). ***$p < .001$ (two-tailed).

TABLE 3. Results of Stepwise Multiple Regression Analysis for the Prediction of Extroversion Scores

Predictor variables	R	R^2	ΔR^2	F	p
One predictor variable					
N1 amplitude	.112	.083			
S-LRP latency	.127	.116			
LRP-R latency	.380	.145	.145	5.07	.032
Two predictor variables					
N1 amplitude, LRP-R latency	.499	.249	.137	5.28	.029
LRP-R latency, S-LRP latency	.504	.254			
N1 amplitude, S-LRP latency	.537	.289			
Three predictor variables					
LRP-R latency, S-LRP latency, N1 amplitude	.635	.403	.150	7.02	.013

Note. See Table 1 for abbreviations. The correlation coefficients between N1 amplitude, S-LRP latency, and R-LRP latency are given in Table 2 (in bold type). Beginning with the first line, predictors and combinations of predictors are ordered by increasing amount of total variance (R^2). Changes in R^2 (ΔR^2), F, and p are denoted for the best-fitting model.

analyze stimuli faster than extroverts can. Those findings, however, do not falsify Brebner and Cooper's (1985) model but instead allow for an extension in which the specific task requirements are taken into account.

Unlike the experimental protocol we used in the present experiment, Rammsayer and Stahl (2004) used an auditory pitch-discrimination task (800 Hz, 1200 Hz, and white noise) and failed to demonstrate extroversion-related differences

in S-LRP latencies. There were two major differences between their study and the present one. On the one hand, both we and Stahl and Rammsayer (2004) used visual stimuli to establish an effect of extroversion on S-LRP latency. That modality may be crucial for evoking extroversion-related differences in stimulus analysis. On the other hand, discrimination of pitch represents an almost automatic process located at a subcortical level of the central nervous system (Chatterjee & Zwislocki, 1997; Moller, 1999), whereas discrimination of visually presented letters represents a more complex cognitive task. From that perspective, extroversion-related differences in speed of premotor information processing appear to become evident only at relatively high levels of cognitive task demands rather than with purely sensory tasks. Nevertheless, on the basis of the existing findings we cannot finally decide whether task difficulty, stimulus modality, or still another variable was responsible for the studies' differences between the effects of extroversion on the duration of stimulus analysis. Future studies in which researchers use an experimental design that independently varies those factors are necessary.

Extroverts' shorter LRP-R latencies and the reliable negative correlational relationship between individual extroversion scores and LRP-R latencies point to faster motor processing in extroverts than in introverts. With an auditory go/no-go task, Rammsayer and Stahl (2004) also found shorter LRP-R latencies for extroverts than for introverts, whereas Stahl and Rammsayer (2004) found no evidence of extroversion-related LRP-R differences with a visual two-choice reaction time task without a no-go condition. Thus, for extroversion-related differences in LRP-R latency to become evident, the no-go condition appears to be a *conditio sine qua non*. That finding supports Brebner and Cooper (1985), who suggested that there is faster response organization in extroverts than introverts. The present LRP-R results leave unclear whether the extroversion-related difference in speed of motor processing was primarily the result of an accelerating effect of the go/no-go task on extroverts, a decelerating effect on introverts, or a combination of both. Given the higher rate of no-go errors in extroverts, a higher level of response excitation (cf. Brebner, 1985) brought about by the go/no-go condition could account for an accelerating effect on extroverts' speed of motor processing.

In future research, the investigation of individual differences in response strategies will be an important issue. Although in the present study we failed to obtain a group difference in response speed, we did find a reliable difference in response accuracy for no-go responses. Compared with extroverts, introverts may reduce their motor response speed to reduce their response errors. It is known from several studies (e.g., Brebner & Flavel, 1978; Helmers et al., 1997; Rammsayer & Stahl, 2004; Socan & Bucik, 1998) that introverts prefer accuracy over speed, whereas extroverts tend to respond faster, disregarding accuracy. Therefore, differential speed–accuracy tradeoff in extroverts and introverts may have been relevant for the present results. Although Socan and Bucik found no behavioral evidence for that propostion, extroversion might be accompanied by differences in speed–accuracy tradeoffs, if researchers separately consider the latter for premotor and motor processing. Investigators could reveal such a pattern by using an electrophysiological approach in which the LRP is applied. The results of a recent LRP study could demonstrate an effect of speed–accuracy tradeoff on both premotor and motor processes (Rinkenauer, Osman, Ulrich, Müller-Gethmann, & Mattes, 2004). Most important in the context of the present findings, Rinkenauer et al. observed longer motor processing with an accuracy instruction than with a speed instruction. Thus, an interesting approach would be to investigate extroversion-related differences in premotor and motor speed–accuracy tradeoffs in the experimental task applied by Rinkenauer et al..

Because LRP-R latency includes speed of both central and peripheral motor processing, it is still unclear whether central rather than peripheral aspects of motor processing mediated extroverts' faster motor speed. Therefore, we analyzed EMG latencies in addition to LRP-R latencies. Both the finding of longer mean EMG latencies in introverts and the negative correlational relationship between EMG latencies and individual extroversion scores suggest that the time from onset of the activation of forearm muscles to onset of the key press was shorter in extroverts than in introverts. Those findings point to extroversion-related differences in peripheral motor processes. The finding that MT, but not RT, is faster for extroverts than for introverts provides converging evidence for extroversion-related differences at the peripheral level (Barratt, 1967; Doucet & Stelmack, 1997, 2000; Wickett & Vernon, 2000). Furthermore, Doucet and Stelmack showed that it is the early acceleration phase of MT that accounts for extroverts' faster MT.

However, additional correlations that we computed between LRP-R latencies and individual extroversion scores while we controlled for EMG latency showed that the reliable negative relationship between LRP-R latencies and individual extroversion scores remained virtually unchanged (rs = –.38 and –.39, respectively) after EMG latency was partialed out. That finding supports the view that individual differences in extroversion are related to differences in fundamental motor processes (Doucet & Stelmack, 1997, 2000; Stelmack & Pivik, 1996). The present data even expand that view by showing that central mechanisms associated with motor activation, inherent in RT and thus not reflected by MT, appear to also contribute to behavioral differences between introverts and extroverts. Thus, extroversion-related differences in speed of motor processing appear to originate from both central and peripheral levels of motor processing.

Challenging Brebner and Cooper's (1985) model, LRP analyses supported the concept that central processes associated with stimulus analysis are faster in introverts than in extroverts. Further support for that view was indicated by

introverts' shorter S-LRP latencies. However, some evidence of higher speed of motor processes in extroverts than in introverts emerged, as reflected by introverts' longer LRP-R latencies. That overall pattern of LRP results may account for our inability in the present study and many previous researchers' inability (e.g., Barratt, 1967; Doucet & Stelmack, 1997, 2000; Rammsayer & Stahl, 2004; Stahl & Rammsayer, 2004; Wickett & Vernon, 2000) to demonstrate extroversion-related differences in RT. Because RT represents a composite measure of premotor and motor speed, extroversion-related differences in processing speed could be present at distinct stages of sensorimotor information processing. However, extroverts' faster motor processes may compensate for their lower speed of stimulus analysis. As a result, no differences in RT between introverts and extroverts should become evident at the behavioral level. That kind of compensation may account for the common finding of almost identical RTs for introverts and extroverts.

To identify the most powerful variables for the prediction of individual extroversion scores, we subjected all behavioral and psychophysiological measures obtained in the present study to stepwise multiple regression analysis. As the most powerful single predictor variable, LRP-R latency accounted for 14% of total variance in variability of extroversion. When combined, two variables, S-LRP latency and N1-amplitude, explained the largest portion of total variance (28.9%). The combination of LRP-R latency, S-LRP latency, and N1 amplitude resulted in another significant increase in explained variance. Thus, those three variables together predicted more than 40% of extroverts' overall variability. Inclusion of any further predictor variable did not result in a statistically significant increase in explained variance. That outcome suggests that speed of premotor and motor processing, as reflected by S-LRP and LRP-R latencies, respectively, and aspects of early sensory processing, as reflected by N1 amplitude, represent the major constituent elements of extroversion-related differences in sensorimotor processing. A cautionary note concerns the fact that our study sample consisted of only women, which prevents us from generalizing the findings to men. Further studies with equal proportions of men and women participants are required.

In sum, although behavioral measures of processing speed have often failed to reveal extroversion-related differences, the present study provided further evidence for faster premotor processing in introverts and faster central and peripheral motor processing in extroverts. Those findings support a speed compensation hypothesis of extroversion. To further elucidate extroversion-related differences at distinct stages of premotor and motor processing, future researchers should apply experimental variations that specifically affect distinct components of premotor and motor processing.

ACKNOWLEDGMENT

Jutta Stahl is indebted to Henning Gibbons for helpful comments and contributions he made on earlier drafts of the manuscript and for his understanding.

Biographical Notes

Jutta Stahl's major research interests are psychophysiological facets of individual differences, motor processing, and methodical aspects of event-related potentials. She teaches personality, intelligence, psychological assessment, and advanced statistics.

Thomas Rammsayer is interested in research on intelligence, personality, psychopharmacology, and time perception. He teaches differential psychology and psychological assessment.

REFERENCES

Barratt, E. S. (1967). Perceptual-motor performance related to impulsiveness and anxiety. *Perceptual and Motor Skills, 25,* 485–492.

Brebner, J. (1985). Personality theory and movement. In B. D. Kirkcaldy (Ed.), *Individual differences in movement* (pp. 27–41). Lancaster, England: MTP Press.

Brebner, J., & Cooper, C. (1974). The effect of a low rate of regular signals upon the reaction times of introverts and extroverts. *Journal of Research in Personality, 8,* 263–276.

Brebner, J., & Cooper, C. (1978). Stimulus- or response-induced excitation: A comparison of the behavior of introverts and extroverts. *Journal of Research in Personality, 12,* 306–311.

Brebner, J., & Cooper, C. (1985). A proposed unified model of extroversion. In J. T. Spence & C. E. Izard (Eds.), *Motivation, emotion, and personality* (pp. 219–229): New York: Elsevier Science.

Brebner, J., & Flavel, R. (1978). The effect of catch-trials on speed and accuracy among introverts and extroverts in a simple RT task. *British Journal of Psychology, 69,* 9–15.

Brocke, B., & Liepmann, D. (1985). Eysencks Theorie der situativen Erregung: Einige Befunde zu einer interaktionistischen Ergänzung der Aktivierungstheorie der Persönlichkeit [Eysenck's theory of situational arousal: Some evidence for an interactional extension of the arousal theory of personality]. *Zeitschrift für Differentielle und Diagnostische Psychologie, 6,* 19–31.

Bullock, W. A., & Gilliland, K. (1993). Eysenck's arousal theory of introversion-extroversion: A converging measures investigation. *Journal of Personality and Social Psychology, 64,* 113–123.

Cahill, J. M., & Polich, J. (1992). P300, probability, and introverted/extroverted personality types. *Biological Psychology, 33,* 23–35.

Chatterjee, M., & Zwislocki, J. J. (1997). Cochlear mechanisms of frequency and intensity coding. I. The place code for pitch. *Hearing Research, 111,* 65–75.

Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd ed.). Hillsdale, NJ: Erlbaum.

Coles, M. G. (1989). Modern mind-brain reading: Psychophysiology, physiology, and cognition. *Psychophysiology, 26,* 251–269.

Daruna, J. H., Karrer, R., & Rosen, A. J. (1985). Introversion, attention and the late positive component of event-related potentials. *Biological Psychology, 20,* 249–259.

De Pascalis, V. (2004). On the psychophysiology of extroversion. In R. M. Stelmack (Ed.), *On the psychobiology of personality* (pp. 295–328). Amsterdam: Elsevier.

Ditraglia, G., & Polich, J. (1991). P300 and introverted/extroverted personality types. *Psychophysiology, 28,* 177–184.

Doucet, C., & Stelmack, R. M. (1997). Movement time differentiates extroverts from introverts. *Personality and Individual Differences, 23,* 775–786.

Doucet, C., & Stelmack, R. M. (2000). An event-related potential analysis of extroversion and individual differences in cognitive processing speed and response execution. *Journal of Personality and Social Psychology, 78,* 956–964.

Eysenck, H. J. (1967). *The biological basis of personality*. Springfield, IL: Charles C. Thomas.

Eysenck, H. J., & Eysenck, S. B. G. (1991). *Manual of the Eysenck Personality Scales (EPS Adult).* London, England: Hodder and Stoughton.

Frewer, L. J., & Hindmarch, I. (1988). The effect of time of day, age, and anxiety on a choice reaction task. In I. Hindmarch, B. Aufdembrinke, & H. Ott (Eds.), *Psychopharmacology and reaction time* (pp. 103–114). Chichester, England: Wiley.

Fridlund, A. J., & Cacioppo, J. T. (1986). Guidelines for human electromyographic research. *Psychophysiology, 23,* 567–589.

Giray, M., & Ulrich, R. (1993). Motor coactivation revealed by response force in divided and focused attention. *Journal of Experimental Psychology: Human Perception and Performance, 19,* 1278–1291.

Gratton, G., Coles, M. G. H., Sirevaag, E. J., Eriksen, C. W., & Donchin, E. (1988). Pre- and poststimulus activation of response channels: A psychophysical analysis. *Journal of Experimental Psychology: Human Perception and Performance, 14,* 331–344.

Gurrera, R. J., O'Donnell, B. F., Nestor, P. G., Gainski, J., & McCarley, R. W. (2001). The P3 auditory event-related brain potential indexes major personality traits. *Biological Psychiatry, 49,* 922–929.

Helmers, K. F., Young, S. N., & Pihl, R. O. (1997). Extroversion and behavioral impulsivity. *Personality and Individual Differences, 23,* 441–452.

Hillyard, S. A., Hink, R. F., Schwent, V. L., & Picton, T. W. (1973, October 12). Electrical signs of selective attention in the human brain. *Science, 182*(4108), 177–179.

Jasper, H. H. (1958). The ten twenty electrode system of the International Federation. *Electroencephalography and Clinical Neurophysiology, 10,* 371–375.

Jensen, A. R., & Munro, E. (1979). Reaction time, movement time, and intelligence. *Intelligence, 3,* 121–126.

Keuss, P. J., & Orlebeke, J. F. (1977). Transmarginal inhibition in a reaction time task as a function of extroversion and neuroticism. *Acta Psychologica, 41,* 139–150.

Kutas, M., McCarthy, G., & Donchin, E. (1977, August 19). Augmenting mental chronometry: The P300 as a measure of stimulus evaluation time. *Science, 197*(4305), 792–795.

Leuthold, H., & Sommer, W. (1998). Postperceptual effects and P300 latency. *Psychophysiology, 35,* 34–46.

Lynn, R., & Martin, T. (1997). Gender differences in extroversion, neuroticism, and psychoticism in 37 nations. *The Journal of Social Psychology, 137,* 369–373.

Miller, J., Patterson, T., & Ulrich, R. (1998). Jackknife-based method for measuring LRP onset latency differences. *Psychophysiology, 35,* 99–115.

Moller, A. R. (1999). Review of the roles of temporal and place coding of frequency in speech discrimination. *Acta Otolaryngology, 119,* 424–430.

Osman, A., & Moore, C. M. (1993). The locus of dual-task interference: Psychological refractory effects on movement-related brain potentials. *Journal of Experimental Psychology: Human Perception and Performance, 19,* 1–21.

Osman, A., Moore, C. M., & Ulrich, R. (1995). Bisecting RT with lateralized readiness potentials: Precue effects after LRP onset. *Acta Psychologica, 90,* 111–127.

Pritchard, W. S. (1981). Psychophysiology of P300. *Psychological Bulletin, 89,* 506–540.

Rammsayer, T., & Stahl, J. (2004). Extroversion-related differences in response organization: Evidence from lateralized readiness potentials. *Biological Psychology, 66,* 35–49.

Rinkenauer, G., Osman, A., Ulrich, R., Müller-Gethmann, H., & Mattes, S. (2004). On the locus of speed-accuracy trade-off in reaction time: Inferences from the lateralized readiness potential. *Journal of Experimental Psychology: General, 133,* 261–282.

Ruch, W. (1999). Die revidierte Fassung des Eysenck Personality Questionnaire und die Konstruktion des deutschen EPQ-R bzw. EPQ-RK [The Eysenck Personality Questionnaire-Revised and the construction of German standard and short versions (EPQ-R and EPQ-RK)]. *Zeitschrift für Differentielle und Diagnostische Psychologie, 20,* 1–24.

Sanders, A. F. (1998). *Elements of human performance: Reaction processes and attention in human skill*. London: Erlbaum.

Smid, H. G. O. M., Mulder, G., & Mulder, L. J. M. (1987). The continuous flow model revisited: Perceptual and central motor aspects. In R. Johnson, Jr., J. W. Rohrbaugh, & R. Parasuraman (Eds.), *Current trends in event-related potential research, EEG Suppl. 40* (pp. 270–278). Amsterdam: Elsevier.

Smith, E. E. (1968). Choice reaction time: An analysis of the major theoretical positions. *Psychological Bulletin, 69,* 77–110.

Socan, G., & Bucik, V. (1998). Relationship between speed of information-processing and two major personality dimensions: Extroversion and neuroticism. *Personality and Individual Differences, 25,* 35–48.

Stahl, J., & Gibbons, H. (2004). The application of jackknife-based onset detection of lateralized readiness potential in correlative approaches. *Psychophysiology, 41,* 845–860.

Stahl, J., & Rammsayer, T. (2004). Differences in the transmission of sensory input into motor output between introverts and extroverts: Behavioral and psychophysiological analyses. *Brain and Cognition, 56,* 293–303.

Stelmack, R. M. (1985). Personality and motor activity: A psychophysiological perspective. In B. D. Kirkcaldy (Ed.), *Individual differences in movement* (pp. 192–213). Lancaster, England: MTP Press.

Stelmack, R. M. (1997). The psychophysics and psychophysiology of extroversion and arousal. In H. Nyborg (Ed.), *The scientific study of human nature: A tribute to Hans Eysenck at eighty* (pp. 388–403). Oxford, England: Pergamon Press.

Stelmack, R. M., & Houlihan, M. (1995). In D. H. Saklofske & M. Zeidner (Eds.), *International handbook of personality and intelligence. Perspectives on individual differences* (pp. 349–365). New York: Plenum Press.

Stelmack, R. M., Houlihan, M., & McGarry-Roberts, P. A. (1993). Personality, reaction time, and event-related potentials. *Journal of Personality and Social Psychology, 65,* 399–409.

Stelmack, R. M., & Pivik, R. T. (1996). Extroversion and the effect of exercise on spinal motoneuronal excitability. *Personality & Individual Difference, 21,* 69–76.

Stenberg, G. (1994). Extroversion and the P300 in a visual classification task. *Personality and Individual Differences, 16,* 543–560.

Verleger, R. (1997). On the utility of P3 latency as an index of mental chronometry. *Psychophysiology, 34,* 131–156.

Wickett, J. C., & Vernon, P. A. (2000). Replicating the movement time–extroversion link. With a little help from IQ. *Personality and Individual Differences, 28,* 205–215.

Submitted November 23, 2006
Revised February 16, 2007